

B.Tech. Degree VI Semester Examination, April 2008

ME 605 CONTROL SYSTEMS ENGINEERING

(1999 Scheme)

Time: 3 Hours

Maximum Marks: 100

- I
- With the aid of simple example bring out the difference between an open loop system and a closed loop system. (5)
 - Find the inverse Laplace transform of: (5)

$$\text{i) } F(s) = \frac{5S+3}{(S+1)(S+2)(S+3)}$$

$$\text{ii) } F(s) = \frac{2S+12}{S^2+2S+5} \quad (15)$$

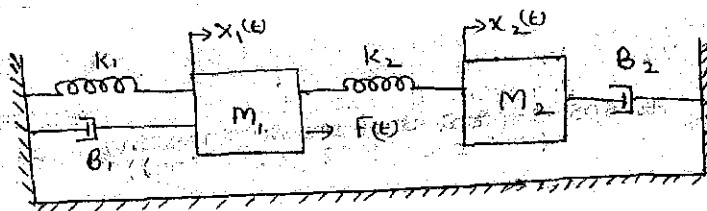
OR

- II
- Draw the response of a 2nd order underdamped system to a unit step input and mark all the time domain specifications on it. (8)
 - Solve the following differential equation by Laplace transform method. (8)

$$\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 2x = 5u$$

where u is a unit step function. The initial conditions are $X(0) = -1$ and $\frac{dx}{dt}$ at $t=0$ equals 2. (12)

- III
- For the mechanical system shown in figure derive the transfer function $\frac{X_2(s)}{F(s)}$. Also draw the electrical analogous circuit using force-current analogy. (20)



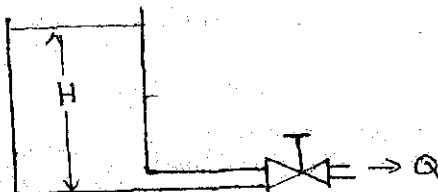
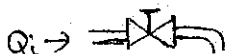
OR

- IV
- Derive the transfer function of an armature controlled D.C motor. (8)
 - In the liquid level system of figure assume that the out flow rate Q m³/sec through the outflow valve is related to the head H m by (8)

$$Q = K\sqrt{H} = 0.01\sqrt{H}$$

Assume also that when the inflow rate Q_i is 0.015 m³/sec the head stays constant. For $t < 0$ the system is at steady state ($Q_i = 0.015$ m³/sec).

At $t = 0$ the inflow valve is closed and so there is no inflow for $t \geq 0$. Find the time necessary to empty the tank to half the original head. The capacitance C of the tank is 2 m².



(12)

- a) Explain Routh Hurwitz criterion by considering a suitable example. (5)
 b) The open loop transfer function of a unity feed back system is given by

$$G(s) = \frac{K}{S(S+3)(S^2+S+1)}.$$

Determine the value of K that will be cause sustained oscillations in the closed loop system. Also find the oscillation frequency. (15)

OR

- a) Explain Liapunov's stability criterion applied to linear systems. (5)
 b) The forward path transfer function $G(s)$ of a unity feedback system

$$G(s) = \frac{K}{s(1+0.01s)(1+0.05s)}.$$

Draw the Bode plot and determine the gain margin and phase margin for $k = 50$. (15)

- a) State and explain sampling theorem. (5)
 b) Find Z-transform of the sequences given below:

i) $f(k) = 1, 1, 1, \dots$ for $k = 0, 1, 2, \dots$

ii) $f(k) = \begin{cases} 1 & \text{for } k = 0, 2, 4, 6, \dots \\ -1 & \text{for } k = 1, 3, 5, 7, \dots \end{cases}$ (15)

OR

- a) Solve the difference equation given below using z - transform:
 $X(k+2) = x(k+1) + 0.1x(k) = U(k)$
 $X(0) = 0$ and $x(1) = 0$
 $U(k)$ denotes the sampled version of unit step. (10)
- b) Prepare the Jury's table for the characteristic polynomial $F(2) = 2z^3 + 5z^2 + z + 8$ and deduce stability. (10)

Write short notes on:

- i) Hydraulic controllers
 ii) PID controller
 iii) Pneumatic controllers
 iv) DC servo motors (20)

OR

Write short notes on:

- i) Tacho generators
 ii) Stepper motors
 iii) AC servo motors
 iv) Synchros (20)
